

let's handheld controller. This system provides the same satisfaction as the present omnipod system, such as a non-stop dosage of insulin and refrains from delaying insulin dosage during exercising or other physical activity.

Researchers are now focusing on the Internet of Things (IoT) devices in order to implement them in the healthcare industry. OpenAPS (artificial pancreas system) initiative was built for a diabetic patient to have a smarter way of blood glucose monitoring and controlling the levels rather than waiting for a new system. Patients are able to feed data of CGM in order to give information to software about glucose levels.

With OpenAPS, a piece of hardware is used that communicates with the pump to get past values of dosage and adjust the required insulin supply. Many patients are having the benefits of software by downloading it online. OpenAPS has built a subdermal system that communicates with a smartphone to provide glucose level information and a vibrating module (alarms) that alerts diabetic patients when glucose levels are above or below the threshold values.

FDA also approves Medtronic AP for diabetic patients over fourteen years. The device named MiniMed 670G is referred to as a hybrid closed-loop system that monitors glucose levels after every five minutes and administers insulin when needed. The system is 'first-of-its-kind technology' that combines insulin pumps and glucose monitors in one device.

Now FDA has accepted to cover 7 to 13-year-olds as well. There is advancement in T1D with the development of Zynquist (sotagliflozin) by Sanofi and Lexicon Pharmaceuticals. It is the first oral antidiabetes medication for T1D patients that inhibits SGLT-1 and SGLT-2 (sodium-glucose cotransporter types 1 and 2). SGLT1 manages the glucose absorption in the gastrointestinal tract, whereas SGLT2 is responsible for glucose reabsorption by the kidney.

Zynquist provides important benefits to adults with T1D by reducing blood sugar reabsorption in the kidneys through SGLT2 inhibition and delaying blood sugar absorption through local SGLT1 inhibition in the intestinal tract.

Researchers from Canada are working on doubling up hormone, which can increase the time spent in the

target glucose range to 86 per cent, according to trials on AP. The team delivered two hormones, that is, pramlintide alongside insulin to have better glucose management after diet for T1D patients. The pancreas makes the hormone amylin after meals to manage glucose levels, and pramlintide is a man-made version of amylin. People infected with T1D disease do not have usual amylin release, and so this dual-hormone AP system is used to alter glucose levels.

The worldwide AP market has been divided based on the following factors:

By the method of type. It contains systems like threshold suspend device system, non-threshold suspend device system, control-to-target system, and control-to-range system. When the patient is not responding to threshold alarms, these devices take an immediate control action.

By mode of treatment type. It involves a hybrid closed loop, bi-hormonal, and insulin.

By mode of end-use. It includes clinics, research, hospitals, and others.

Current status in India

As per the International Diabetes Foundation, India has the highest number of diabetic patients in the world. This disease has its impact on more than fifty million Indians and kills about one million patients every year. This big frequency is a result of the sluggish lifestyle and consumption of high-calorie diet by the growing middle class. So, in order to combat diabetes, various scientific institutions in India are doing efforts to develop AP.

Indian Institute of Science (IISc), Bengaluru, has developed an artificial pancreas to treat T1D. The scientists and researchers are working together with specialists at the MS Ramaiah Medical Hospital to design the pancreas system. The Ministry of Human Resource Development (MHRD) and the Indian Council of Medical Research (ICMR) have given 36 million rupees for the project.

The Narayana Multispecialty Hospital in Hyderabad is the first hospital that provides artificial pancreas treatment. Each session of treatment procedure costs around 4,000 rupees, and

a minimum of ten sessions are required to see improvement.

IIT Mumbai has developed a bioartificial pancreas that can be implanted in diabetic patients to treat diabetes. It has been successfully tested on mice and pigs. The bioartificial pancreas device has been developed utilising the polymer-based hollow fibre, which will be fitted near the pancreas in the abdominal muscles. It can develop cells that secrete insulin.

Opportunities for electronics and computer science engineers

IoT has transformed the field of electronics engineering. IoT has various applications, especially in the healthcare industry. With the help of IoT, engineers can send real-time data of concentrations of blood glucose to a number of devices they want to connect, such as from an Android phone to an iPad or smartwatch that makes day-to-day monitoring of diabetes easy.

The interest and development in AI have been increasing in the healthcare industry over the past decades. Engineers can apply various methods of AI in the management of diabetes and predict the risk of diabetic complications. Machine learning (ML) methods used in food recognition applications can help patients with T1D to have more correct choices of food and insulin management decisions. By using deep learning methods, engineers are able to have a more accurate estimation of blood glucose levels and detect the stage of hypoglycemia.

Conclusion

The innovation of AP has proved to be a boon for diabetic patients and opened a new field of research for the electronics and computer engineering domain. This medical device has the potential to bring peace of mind to more than one million people infected with T1D. We need to make this device available to all the people who are infected with diabetes and make sure that this technology is flexible enough to use in the real world to achieve the best outcomes.

Other thrilling advances include the development of autonomous systems that not only monitor insulin but also glucagon (which raises blood sugar level) in order to have complete control over sugar levels. **EFY**